

PDE

Partial differential eqn: A differential equation having two or more independent variables and partial differential co-efficients of dependent variable with respect to the independent variables is called partial differential equation.

The partial differential eqn $\phi(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}) = 0$ can be written as $\phi(x, y, z, p, q) = 0$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$

Order of PDE: The order of a PDE equation is that of the derivative of highest order in the equation.

Degree of PDE: The degree of a PDE is the greatest exponent of the highest order.

Example: ① $\frac{\partial^4 z}{\partial x^4} + \left(\frac{\partial z}{\partial y}\right)^2 = z \rightarrow \text{Order} = 4, \text{Degree} = 1$

② $\left(\frac{\partial^3 z}{\partial x^3}\right)^2 + x^4 \left(\frac{\partial z}{\partial y}\right)^3 = 2 \rightarrow \text{Order} = 3, \text{Degree} = 2$

③ $x^3 p + y^2 q = 2z \rightarrow \text{Order} = 1, \text{Degree} = 1$

④ $p^2 + q^2 = m \rightarrow \text{Order} = 1, \text{Degree} = 2$

Linear and non-linear PDE:

The PDE is called linear if it is of first degree in the dependent variable and its partial derivatives and product of the dependent variable and its partial derivative must be absent otherwise it is called a non-linear PDE.

Example: ① $x^2 p + y^2 q = z$ is a linear PDE.

② $p^2 + q^2 = 2$ is a non linear PDE.

Question: Form a partial differential eqn from $z = A e^{pt} \sin px$ by eliminating arbitrary constants.

Soln: Given that $z = A e^{pt} \sin px \dots \text{--- (1)}$

Differentiating (1) partially with respect to x and it

$$\frac{\partial z}{\partial x} = A e^{pt} \cos px \cdot p \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial t} = A p e^{pt} \sin px \quad \text{--- (3)}$$

Again differentiating (2) and (3) partially with respect to x and t respectively.

$$\frac{\partial^2 z}{\partial x^2} = -Ap^2 e^{Pt} \sin px \quad \text{--- (4)}$$

$$\frac{\partial^2 z}{\partial t^2} = Ap^2 e^{Pt} \sin px \quad \text{--- (5)}$$

Adding equation (4) and (5) we get,

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$$

This is the partial differential equation.

Question 2 Form a partial differential equation by eliminating arbitrary constants from $z = (x-a)^2 + (y-b)^2$

Soln: Given that $z = (x-a)^2 + (y-b)^2$ --- (1)

Differentiating (1) partially with respect to x and y we get,

$$\frac{\partial z}{\partial x} = 2(x-a) \quad \& \quad \frac{\partial z}{\partial y} = 2(y-b) \quad \text{--- (2)}$$

Squaring and adding eqn (2) and (3) we get,

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(x-a)^2 + 4(y-b)^2$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4[(x-a)^2 + (y-b)^2]$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4z \quad [\text{using (1)}]$$

which is the required partial differential eqn.

Linear PDE: If p, q, z are linear then

$\Phi(x, y, z, p, q) = 0$ is called linear PDE of order one. The general form of the equation is

$$P(x, y)p + Q(x, y)q = R(x, y, z) + S(x, y)$$

Ex: $2xy^2p + x^2yq = xy^2z + xy$

Lagrange's Equation: A linear PDE of the form $p + q = R$, where p, q and R are functions of x, y and z is called Lagrange's equation.

Ex: $a(p + q) = z$

Working procedure for solving Lagrange's equation:

(i) Make the eqn in the standard form

$$p + q = R$$

(ii) Find Lagrange's auxiliary equation $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{R}$

(iii) Find two linear independent solutions of $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{R}$ using well known methods.

(iv) If $u(x, y, z) = c$ and $v(x, y, z) = c_1$ are two solns then the general solution is $\Phi(u, v) = 0$ or $u = \Phi(v)$ or $v = \Phi(u)$ where Φ is an arbitrary function.

Question 1 Solve the following PDE of Lagrange's form

(i) $x^2 p + y^2 q = z^2$

(ii) $p \cdot \tan x + q \cdot \tan y = \tan z$

(iii) $xy p + y^2 q = zxy - 2x$

Soln (i) Given $x^2 p + y^2 q = z^2$ --- (1)

Here $P = x^2$, $Q = y^2$, $R = z^2$

Lagrange's auxiliary eqn is

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$$

Taking 1st and 2nd term

$$\int \frac{dx}{x^2} = \int \frac{dy}{y^2}$$

$$\Rightarrow -\frac{1}{x} = -\frac{1}{y} + c$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = -c$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = c_1$$

Taking 2nd and 3rd term $\int \frac{dy}{y^2} = \int \frac{dz}{z^2}$

$$\Rightarrow -\frac{1}{y} = -\frac{1}{z} + c$$

$$\Rightarrow \frac{1}{y} - \frac{1}{z} = -e$$

$$\Rightarrow \frac{1}{y} - \frac{1}{z} = e^2$$

Hence $\phi\left(\frac{1}{x} - \frac{1}{y}, \frac{1}{y} - \frac{1}{z}\right) = 0$ is the required soln.

Q. Solve $p - 2q = 3x^2 \sin(y + 2x)$

Soln: Given $p - 2q = 3x^2 \sin(y + 2x) \dots (1)$

Here $P = 1, Q = -2, R = 3x^2 \sin(y + 2x)$

Lagrange's auxiliary equation is

$$\frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y + 2x)} \dots (2)$$

Taking first two terms

$$\frac{dx}{1} = \frac{dy}{-2}$$

$$\int dx = \int -\frac{dy}{2}$$

$$\Rightarrow \int 2 dx = \int -dy$$

$$\Rightarrow 2x = -y + e$$

$$\Rightarrow 2x + y = e \dots (3)$$

Taking 2nd and 3rd term

$$\frac{dx}{1} = \frac{dz}{3x^2 \sin(y + 2x)}$$

$$\int dz = \int 3x^2 \sin(y + 2x) dx$$

$$\int dz = \int 3x^2 \sin a dx$$

let, $y + 2x = a$

$$\Rightarrow z = \frac{2x^3}{3} \sin a + c$$

$$\Rightarrow z = x^3 \sin a + c$$

$$\Rightarrow x^3 \sin a - z = c$$

$$\Rightarrow x^3 \sin(2x+y) - z = c$$

Hence $\phi(2x+y, x^3 \sin(2x+y)) = 0$ is the required solution.

Ex Solve $zp = -x$

Soln. The given equation

$$zp = -x \quad \dots (1)$$

Comparing with the standard form $Pp + Qq = R$

$$P = z, \quad Q = 0, \quad R = -x$$

Lagrange's Auxiliary equation for (1)

$$\frac{dx}{z} = \frac{dy}{0} = \frac{dz}{-x} \quad \dots (2)$$

Taking 1st and 3rd term

$$\frac{dx}{z} = \frac{dz}{-x}$$

$$\Rightarrow \int x dx = - \int z dz$$

$$\Rightarrow \int x dx + \int z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{z^2}{2} = c_1$$

$$\Rightarrow x^2 + z^2 = 2c_1$$

$$\therefore x^2 + z^2 = c_1$$

Again from 2nd and 3rd $\frac{dy}{dz} = \frac{dz}{-x}$

$$x dy = 0$$

$$\Rightarrow dy = 0$$

$$\Rightarrow \int dy = \int 0$$

$$\Rightarrow y = c$$

The required soln is $\phi(x^2 + z^2, y) = 0$

Try yourself:

⑩ Solve $x^2 p + y^2 q + z^2 = 0$ Am: $\phi\left(\frac{1}{x} - \frac{1}{y}, \frac{1}{y} + \frac{1}{z}\right) = 0$

⑪ Solve $\frac{y^2 z}{x} p + x z q = y^2$ Am: $\phi(x^3 y^3, x^2 + z^2) = 0$

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Linear Homogeneous PDE ^{of Higher order} with constant
co-efficient

The general form of n th order linear homogeneous
 PDE with constant co-efficients is

$$(A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D'^n)z = f(x, y)$$

$$\text{OR, } F(D, D')z = f(x, y)$$

$$\text{where } \frac{\partial z}{\partial x} = p = D'z, \frac{\partial z}{\partial y} = q = D''z, \frac{\partial^2 z}{\partial x^2} = r = D'^2 z$$

$$\frac{\partial^2 z}{\partial x \partial y} = s = DD'z, \frac{\partial^2 z}{\partial y^2} = t = D'^2 z$$

The equation $2 \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 29(y-x)$
 can be written as -

$$2r - 5s + 2t = 29(y-x)$$

$$\text{OR, } (2D'^2 - 5DD' + 2D'^2)z = 29(y-x)$$

The General soln will be $z = C.F + P.I$

Q. Finding C.F: For auxiliary equation we have to put $D=m$ and $D'=1$ in $F(D, D')=0$. Hence the A.E is $F(m, 1)=0$

Case I: If $m_1, m_2, \dots, m_n$ are distinct then the C.F = $\phi_1(y+m_1x) + \phi_2(y+m_2x) + \dots + \phi_n(y+m_nx)$

Case II: For repeated roots i.e. $m_1=m_2=\dots=m_n=m_1$ then the C.F = $\phi_1(y+m_1x) + x\phi_2(y+m_1x) + x^2\phi_3(y+m_1x) + \dots + x^{n-1}\phi_n(y+m_1x)$

Case III: (i) If $m = \pm ai$ then the

$$C.F = \phi_1(y+aix) + \phi_2(y-aix)$$

(ii) If $m = a \pm ib$ then the

$$C.F = \phi_1[y+(a+ib)x] + \phi_2[y+(a-ib)x]$$

Q. F(D, D')z=0:

Problem 1: solve $(D^3 - 3D^2D' + 2DD'^2)z=0$

Soln: Given that, $(D^3 - 3D^2D' + 2DD'^2)z=0$ --- (1)

putting $D=m$ and $D'=1$ in eq(1) we get,

Auxiliary equation $m^3 - 3m^2 + 2m = 0$

$$\Rightarrow m(m^2 - 3m + 2) = 0$$

$$\Rightarrow m\{m^2 - 2m - m + 2\} = 0$$

$$\Rightarrow m \{ m(m-2) - 1(m-2) \} = 0$$

$$\Rightarrow m(m-2)(m-1) = 0$$

$$\therefore m = 0, 1, 2$$

The general solution $z = \phi_1(y) + \phi_2(y+x) + \phi_3(y+2x)$

~~Q~~ Solve $2z - 3s + 2t = 0$

Soln: Given that, $2z - 3s + 2t = 0$

$$\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 0$$

$$\Rightarrow (D^2 - 3DD' + 2D'^2)z = 0$$

The A.E is, $m^2 - 3m + 2 = 0$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

The general solution is, $z = \phi_1(y+x) + \phi_2(y+2x)$

~~Q~~ Solve $2z + 5s + 2t = 0$

Soln: Given that, $2z + 5s + 2t = 0$

$$\Rightarrow 2 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 0$$

$$\Rightarrow (2D^2 + 5DD' + 2D'^2)z = 0$$

The A.E is $2m^2 + 5m + 2 = 0$

$$\Rightarrow 2m^2 + 4m + m + 2 = 0$$

$$\Rightarrow 2m(m+2)+1(m+2)=0$$

$$\Rightarrow (m+2)(2m+1)=0$$

$$\therefore m = -2, -\frac{1}{2}$$

The general soln is $z = \phi_1(y-2x) + \phi_2(y-\frac{1}{2}x)$

Solve $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$

Soln: Given that, $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$

A.E is $m^3 - 6m^2 + 11m - 6 = 0$

$$\Rightarrow m^3 - m^2 - 5m^2 + 5m + 6m - 6 = 0$$

$$\Rightarrow m^2(m-1) - 5m(m-1) + 6(m-1) = 0$$

$$\Rightarrow (m-1)(m^2 - 5m + 6) = 0$$

$$\Rightarrow (m-1)(m^2 - 3m - 2m + 6) = 0$$

$$\Rightarrow (m-1)(m-3)(m-2) = 0$$

$$\therefore m = 1, 2, 3$$

The general soln is $z = \phi_1(y+x) + \phi_2(y+2x) + \phi_3(y+3x)$

Solve $\frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0$

$$\Rightarrow (D^4 - D'^4)z = 0$$

The A.E is $m^4 - 1 = 0$

$$(m^2 - 1)(m^2 + 1) = 0$$

$$(m^2 - 1)(m^2 + 1) = 0$$

$$\left. \begin{array}{l} m^2 - 1 = 0 \\ m^2 = 1 \\ m = \pm 1 \end{array} \right\} \begin{array}{l} m^2 + 1 = 0 \\ m^2 = -1 \\ m = \pm i \end{array}$$

$$\therefore m = \pm 1, \pm i$$

$$\text{Then } G.S \text{ is } z = \phi_1(y+x) + \phi_2(y-x) + \phi_3(y+in) + \phi_4(y-in)$$

$$\text{If } \underline{F(D, D')^2 = f(ax+by)}:$$

Case I: IF $F(D, D')$ is a homogeneous function of degree n and $F(a, b) \neq 0$ then

$$P.I = \frac{1}{F(D, D')} \phi(ax+by)$$

$$= \frac{1}{F(a, b)} \int \dots \int \phi(v) dv dv \dots dv \text{ where } v = ax+by$$

$$= \frac{1}{F(a, b)} \int \dots \int \phi(v) dv^n$$

Case II: IF $F(a, b) = 0$ then

$$P.I = \frac{1}{F(D, D')} \phi(ax+by) = x \frac{1}{F_x(D, D')} \phi(ax+by)$$

* IF $F_x(D, D') = 0$ then

$$P.I = x^2 \frac{1}{F_{xx}(D, D')} \phi(ax+by)$$

Question-01 Solve $(2D^2 - 5DD' + 2D'^2)z = 24(y-x)$

Soln: As given that,
 $(2D^2 - 5DD' + 2D'^2)z = 24(y-x)$

A.E $2m^2 - 5m + 2 = 0$

$\Rightarrow 2m^2 - 4m - m + 2 = 0$

$\Rightarrow 2m(m-2) - 1(m-2) = 0$

$\Rightarrow (m-2)(2m-1) = 0$

$\therefore m = \frac{1}{2}, 2$

C.F = $\phi_1(y + \frac{x}{2}) + \phi_2(y + 2x)$

P.I = $\frac{1}{2D^2 - 5DD' + 2D'^2} 24(y-x)$

$= \frac{24}{2(-1)^2 - 5(-1) \cdot 1 + 2(1)^2} \int \int v dv$ where $v = y-x$

$= \frac{24}{9} \int \frac{v^2}{2} dv$

$= \frac{4}{3} \frac{v^3}{3}$

$= \frac{4}{9} v^3$

$= \frac{4}{9} (y-x)^3$

G.S = C.F + P.I

$= \phi_1(y + \frac{x}{2}) + \phi_2(y + 2x) + \frac{4}{9} (y-x)^3$

Question: Solve $12x^2 + 5xy - 2y^2 = (2x+y)^{5/2}$

Soln. A.E. is $m^2 + m - 2 = 0$

$$m = -2, 1$$

$$C.F. = \phi_1(y-2x) + \phi_2(y+x)$$

$$\begin{aligned} \text{Now, P.I.} &= \frac{1}{D^2 + DD' - 2D'^2} (2x+y)^{5/2} \\ &= \frac{1}{4+2-2} \iint v^{5/2} dv dv, \text{ where } v = 2x+y \\ &= \frac{1}{4} \int \frac{v^{7/2}}{7/2} dv \\ &= \frac{1}{6} \frac{v^{9/2}}{9/2} \\ &= \frac{1}{15} (2x+y)^{9/2} \end{aligned}$$

$$\begin{aligned} \text{General soln} &= C.F. + P.I. \\ &= \phi_1(y-2x) + \phi_2(y+x) + \frac{1}{15} (2x+y)^{9/2} \end{aligned}$$

Q1 $f(x,y) = e^{ax+by}$:

Q1 P.I. = $\frac{1}{F(a,b)} e^{ax+by}$; $F(a,b) \neq 0$

Q2 If $F(a,b) = 0$ then

$$P.I. = x \frac{1}{F_x(D,D')} e^{ax+by}$$

Q.1 Solve $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$

Soln: Given that, $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$

A.E $m^2 + 2m + 1 = 0$

$m^2 + m + m + 1 = 0$

$m(m+1) + 1(m+1) = 0$

$(m+1)(m+1) = 0$

$\therefore m = -1, -1$

C.F $z_c = \phi_1(y-x) + x\phi_2(y-x)$

Particular integral $z_p = \frac{1}{D^2 + 2DD' + D'^2} e^{2x+3y}$

$= \frac{1}{(1)^2 + 2 \times 2 \times 3 + (3)^2} e^{2x+3y}$

$= \frac{1}{4 + 12 + 9} e^{2x+3y}$

$= \frac{1}{25} e^{2x+3y}$

General soln: $z = z_c + z_p$

$= \phi_1(y-x) + x\phi_2(y-x) + \frac{1}{25} e^{2x+3y}$

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Question Solve $(D^2 - 5DD' + 6D'^2)z = \sin(4x+y)$

Soln: Given that, $(D^2 - 5DD' + 6D'^2)z = \sin(4x+y)$

$$A.E \quad m^2 - 5m + 6 = 0$$

$$\Rightarrow m^2 - 3m - 2m + 6 = 0$$

$$\Rightarrow m(m-3) - 2(m-3) = 0$$

$$\Rightarrow (m-3)(m-2) = 0$$

$$\therefore m = 2, 3$$

$$C.F \quad z_c = \phi_1(y+2x) + \phi_2(y+3x)$$

$$P.I = \frac{1}{D^2 - 5DD' + 6D'^2} \sin(4x+y)$$

$$= \frac{1}{9^2 - 5 \times 4 \times 1 + 6 \times 1} \int \sin v \, dv \, dv \text{ where } v = 4x+y$$

$$\Rightarrow \frac{1}{16 - 20 + 6} \int (-\cos v) \, dv$$

$$\Rightarrow \frac{-1}{22 - 20} \sin v$$

$$= -\frac{1}{2} \sin(4x+y)$$

General soln $z = C.F. + P.I$

$$= \phi_1(y+2x) + \phi_2(y+3x) - \frac{1}{2} \sin(4x+y)$$

Q.1 Solve $(D^3 - 3D^2D' + 2D'^2D)z = \sin(2x+3y)$

Soln: Given that, $(D^3 - 3D^2D' + 2D'^2D)z = \sin(2x+3y)$

A.E is $m^3 - 3m^2 + 2m = 0$

$$\Rightarrow m(m^2 - 3m + 2) = 0$$

$$\Rightarrow m(m^2 - 2m - m + 2) = 0$$

$$\Rightarrow m \{ m(m-2) - 1(m-2) \} = 0$$

$$\Rightarrow m(m-2)(m-1) = 0$$

$$\therefore m = 0, 1, 2$$

C.F $z_c = \phi_1(y) + \phi_2(y+x) + \phi_3(y+2x)$

P.I = $\frac{1}{D^3 - 3D^2D' + 2D'^2D} \sin(2x+3y)$

$$= \frac{1}{(2)^3 - 3(2)^2(3) + 2(3)^2(2)} \int \int \int \sin v \, dv \, dy \, dz$$

$$= \frac{1}{8 - 36 + 36} \int \int (-\cos v) \, dv \, dy \, dz$$

$$= \frac{-1}{8} \int \sin v \, dv$$

$$= \frac{1}{8} \cos v$$

$$= \frac{1}{8} \cos(2x+3y)$$

General soln is $= \phi_1(y) + \phi_2(y+x) + \phi_3(y+2x) + \frac{1}{8} \cos(2x+3y)$